

CLAYTON, NM, USA

WMO: 723600

Lat: 36.449N

Lon: 103.154W

Elev: 1512

StdP: 84.43

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23051

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-13.2	-10.2	-20.3	0.7	-0.2	-18.1	0.9	1.8	15.8	1.1	14.2	3.1	4.5	30	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	35.5	17.0	34.1	17.0	32.7	17.0	20.1	29.0	19.4	28.5	18.8	28.0	6.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	15.4	21.7	17.0	14.6	21.2	16.3	14.0	21.0	64.2	28.8	61.7	28.3	59.5	27.8	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	12.0	10.9	DB	-17.4	37.6	2.6	1.2	-19.2	38.5	-20.7	39.2	-22.2	39.8	-24.1	40.7	(4)
(5)			WB	-17.9	21.6	2.5	2.4	-19.7	23.4	-21.2	24.8	-22.6	26.1	-24.4	27.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	2.5	3.5	7.6	11.5	16.8	22.2	24.5	23.4	19.6	13.1	7.3	2.3		
	DBStd	9.30	5.68	5.89	5.63	5.13	4.57	3.45	2.61	2.73	3.92	5.04	5.49	5.69		
	HDD10.0	974	238	190	112	44	6	0	0	0	1	27	113	244		
	HDD18.3	2573	492	414	333	208	84	8	1	2	32	171	331	498		
	CDD10.0	2034	4	9	37	88	215	367	448	415	289	124	32	4		
	CDD18.3	592	0	0	1	3	35	125	191	158	70	8	0	0		
	CDH23.3	7183	1	5	33	129	594	1581	2166	1667	811	181	17	0		
	CDH26.7	3295	0	0	3	20	219	788	1122	789	314	38	1	0		
(14) Wind	WSAvg	5.4	5.2	5.5	5.8	6.3	5.8	5.7	5.1	4.7	5.2	5.4	5.4	5.3		
(15) Precipitation	PrecAvg	390	7	8	14	23	50	59	67	69	48	23	14	8		
	PrecMax	653	27	41	58	79	111	126	169	148	133	116	55	28		
	PrecMin	227	0	0	0	1	7	12	6	7	9	0	0	0		
	PrecStd	101	7	9	13	20	28	28	37	39	32	32	16	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	23.3	26.5	29.0	33.7	37.2	37.0	36.1	34.3	30.1	25.4	20.8		
		MCWB	7.5	8.7	10.0	11.3	15.1	15.9	17.2	17.9	15.8	13.2	9.9	7.5		
	2%	DB	18.0	19.7	23.9	26.6	31.3	35.3	35.4	34.4	32.1	27.5	22.9	17.3		
		MCWB	6.3	7.2	9.0	10.7	13.8	16.0	17.4	17.7	15.7	12.7	9.0	6.2		
	5%	DB	15.0	17.2	21.2	24.6	29.2	33.5	34.1	32.9	30.3	25.5	20.0	14.6		
		MCWB	5.2	5.9	8.0	10.1	13.3	15.9	17.7	17.7	15.7	12.1	8.1	5.0		
	10%	DB	11.9	14.0	18.3	22.3	26.9	31.6	32.6	31.3	28.3	22.6	16.9	11.7		
		MCWB	3.5	4.7	7.1	9.5	12.8	16.0	17.8	17.8	15.6	11.1	7.0	3.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	9.0	11.5	14.3	18.5	20.2	21.7	21.2	19.5	17.0	11.7	8.3		
		MCDB	19.3	21.3	21.4	22.6	28.9	30.3	30.6	29.5	26.8	21.9	19.2	18.2		
	2%	WB	6.7	7.6	9.9	12.7	16.4	19.0	20.1	20.1	18.2	15.2	10.1	6.7		
		MCDB	16.8	18.1	21.2	21.3	24.8	28.7	28.8	28.8	26.1	21.4	19.9	16.5		
	5%	WB	5.3	6.3	8.7	11.4	15.3	18.1	19.4	19.4	17.5	13.7	8.8	5.3		
		MCDB	14.4	16.5	19.6	20.9	24.0	27.6	28.6	28.2	25.3	21.0	18.4	13.6		
	10%	WB	3.9	5.0	7.6	10.1	14.2	17.5	18.9	18.7	16.7	12.4	7.5	3.9		
		MCDB	11.3	13.5	17.6	20.2	22.9	26.8	28.4	27.5	24.7	20.6	16.5	11.2		
(35) Mean Daily Temperature Range	MDBR		13.9	14.5	15.2	15.2	15.1	15.4	14.7	13.8	14.0	14.3	14.4	13.2		
	5% DB	MCDBR	17.9	18.9	18.7	18.8	18.0	18.2	16.7	16.3	16.8	17.6	18.1	17.1		
		MCWBR	9.5	9.5	8.4	8.3	7.4	6.3	4.9	5.0	5.6	7.2	8.5	9.0		
	5% WB	MCDBR	17.2	18.0	17.5	16.4	15.9	15.5	14.3	14.0	13.8	14.4	16.5	16.5		
		MCWBR	9.4	9.3	8.5	8.1	7.5	6.2	5.0	5.3	5.3	6.9	8.3	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.241	0.247	0.276	0.298	0.317	0.342	0.364	0.347	0.313	0.280	0.255	0.242		
	taud		2.603	2.610	2.515	2.458	2.465	2.425	2.419	2.461	2.524	2.587	2.574	2.594		
	Ebn at Noon		984	1013	1000	984	964	934	911	923	945	957	954	961		
	Edh at Noon		71	81	98	110	111	115	115	108	96	82	73	68		
(44) All-Sky Solar Radiation	RadAvg		3.05	3.96	5.23	6.42	7.31	7.73	7.41	6.65	5.71	4.38	3.32	2.63		
	RadStd		0.23	0.28	0.40	0.34	0.42	0.34	0.28	0.30	0.29	0.35	0.23	0.15		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)		
(47)	Regional (17 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+80	N/A	(47)	

Nomenclature: See separate page